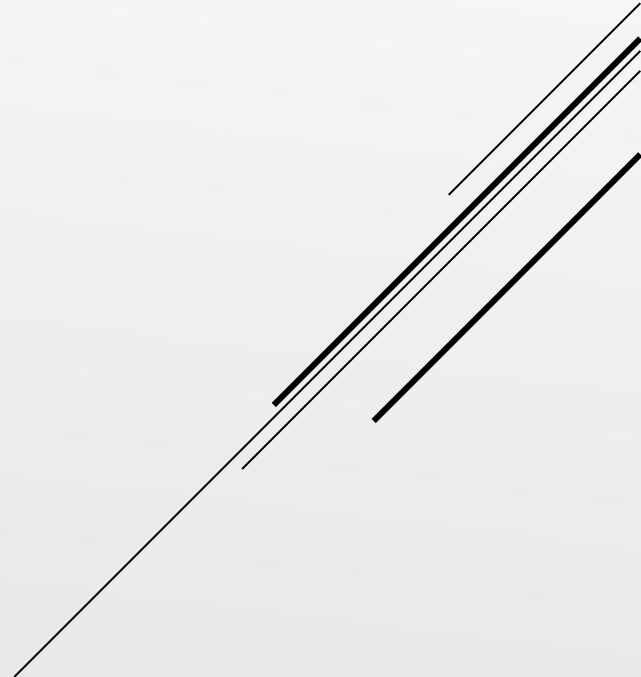


BRINGING A.I. INTO CLAIMS

Exploring the Evolution and Practical Use of Artificial Intelligence in Claims

MS Teams recording: <https://vimeo.com/1178919567/b118fae823?share=copy&fl=sv&fe=ci>



Understanding the History and Evolution of Artificial Intelligence

AI in Claims: Past and Present

AI in Claims: Future

Utilising Generative AI Tools in Claims

Key AI Concepts and Terms for Claims Professionals

AGENDA OVERVIEW



UNDERSTANDING THE HISTORY AND EVOLUTION OF ARTIFICIAL INTELLIGENCE

COMMON MISCONCEPTIONS

AI existed before ChatGPT

Narrow / Weak

Able to perform a narrow task



Strong / General AI

Able to perform any intellectual task that a human can do

AI EXISTED BEFORE CHATGPT

Discriminative AI

Recognises patterns in data to distinguish between different categories or make predictions on input data

Customer behaviour

Fraud detection

Mortality modelling

Lapse modelling

Generative AI (Gen AI)

Creates new, original content by learning from existing data

ChatGPT



BAD DATA CAUSES BAD DECISIONS

Garbage in
=
Garbage out



BAD DATA CAUSES BAD DECISIONS

Biases in the training data will lead to biases in the model

Google DeepMind (2019)

AI system that predicts the presence of acute kidney injury up to two days in advance.

"Female patients comprised 6.38% of patients in the dataset, and model performance was lower for this demographic"

(2022) Performance largely corrected in general population by adding more females to the training set.



BAD DATA CAUSES BAD DECISIONS

AI is not magic

Age

Physical
Activity

BMI

Blood
Pressure

Gender

Shoe
Size

Height

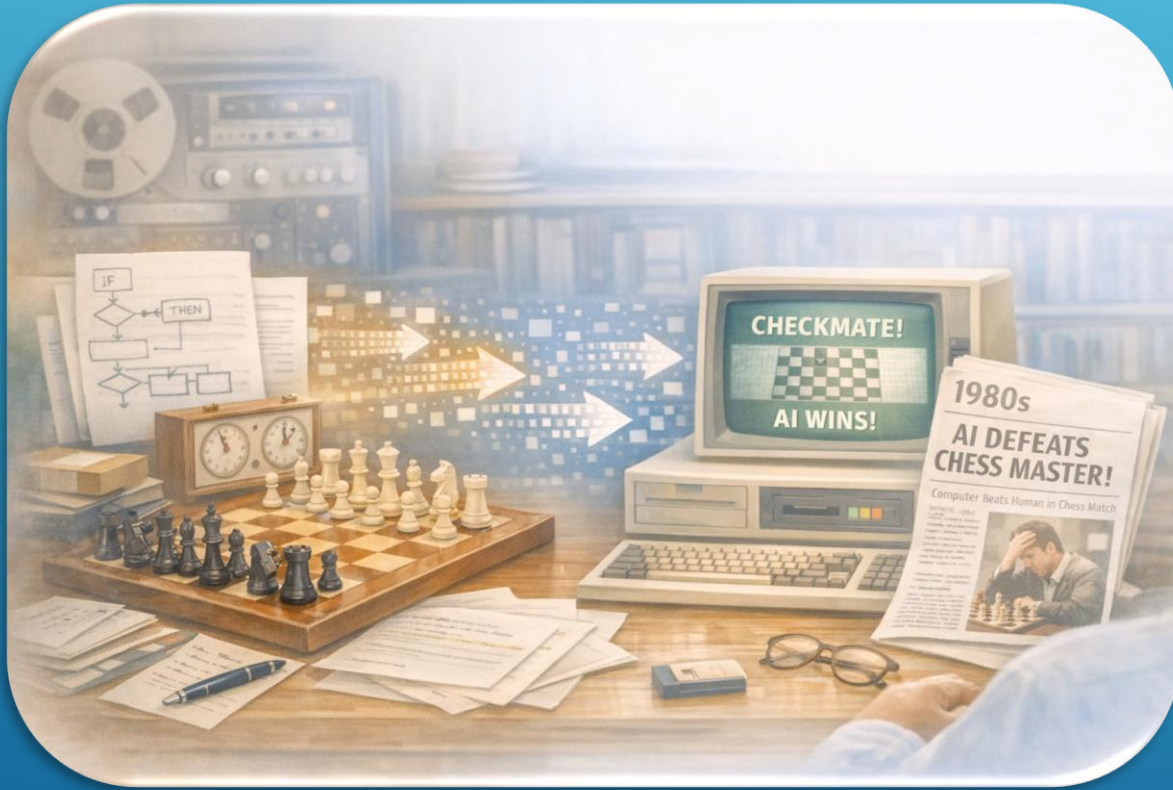
Favourite
Book

Screen
Time

Hair
Length



AI Isn't New: From Early Chess Programs to Today



Early Rule-Based Chess Programs

AI began with simple, rule-based chess programs that followed predefined instructions. These systems didn't learn — they executed fixed strategies programmed by humans

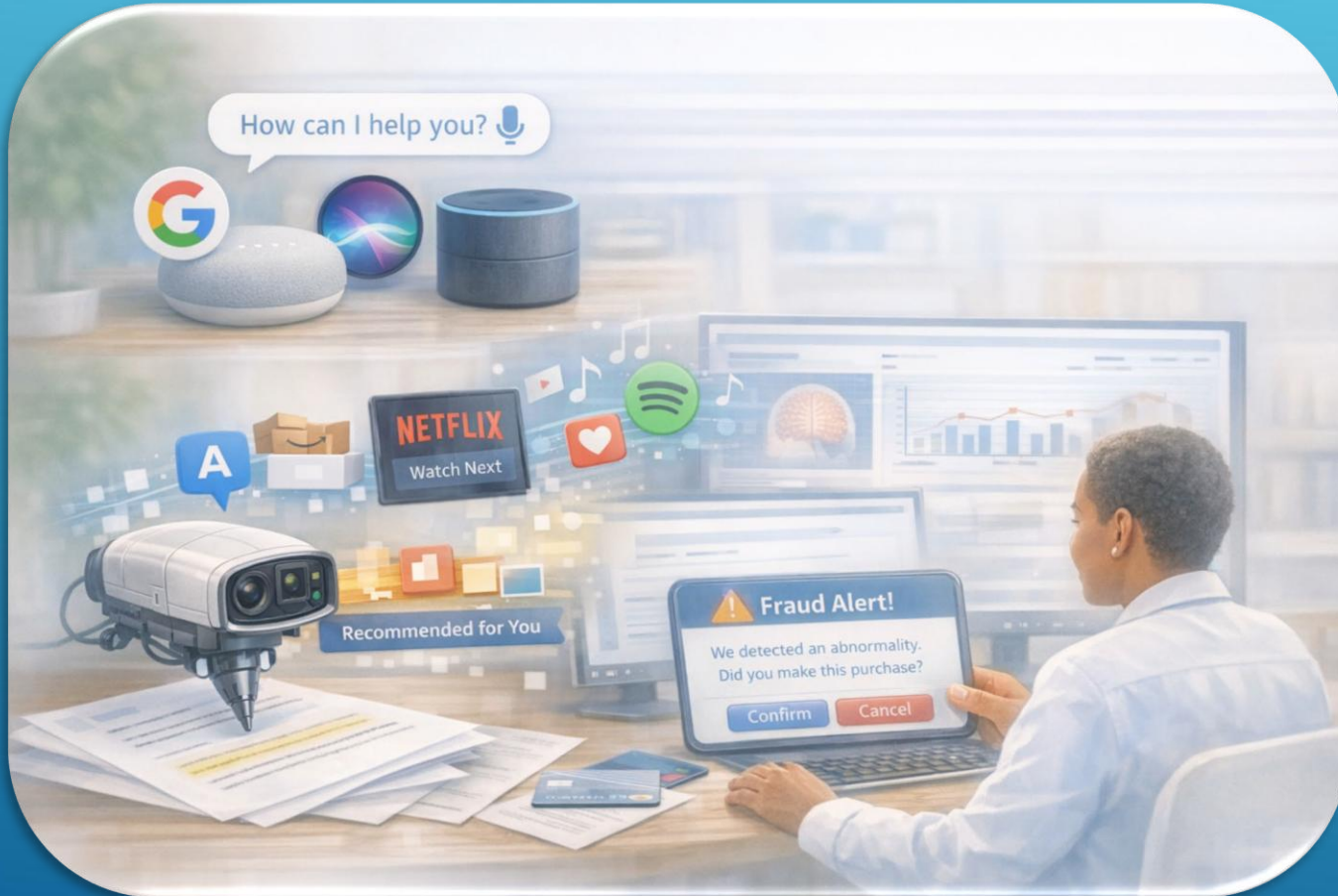
Advances in Computing Power

As computing power increased, AI systems were able to work with much larger datasets and run far more complex calculations. This shift opened the door to more capable and flexible AI models.

Sophisticated Learning Algorithms

Modern AI uses learning algorithms that can recognise patterns in data and produce outputs, such as predictions or recommendations that help support or automate parts of the decision-making process.

Everyday Examples of AI in Our Lives



Virtual Assistants

AI has been part of everyday life for years through familiar virtual assistants like OK Google, Siri, and Alexa, which many people use without even thinking of them as “AI.”

Recommendation Engines

AI quietly shapes what we see online, powering the recommendations on streaming platforms, shopping sites, and social media feeds.

Fraud Detection

Behind the scenes, AI has long been used to help spot unusual activity and keep everyday financial transactions more secure.

Why AI Matters Now, and Its Role in Claims



Breakthroughs in Generative AI

Recent breakthroughs in generative AI — particularly tools that can create text, images, and other content — have brought artificial intelligence back into mainstream attention. These advances have made AI feel more accessible and more visible in everyday work and life.

Large Language Models (LLMs) Impact

LLMs can understand and generate natural language, making it easier to interpret complex information and work with large volumes of text.

Enhancing Claims Assessment

AI tools can support claims teams by helping streamline certain tasks and enhance efficiency in assessment processes.



AI in UK Healthcare What's Happening Now

Brain-mapping research in UK universities is using AI to analyze imaging data, mapping cellular structures to understand how the brain changes after illness or injury, and predict risk of certain diseases (Dementia)

AI-supported prostate cancer screening is improving early detection by spotting subtle patterns in scans for triage before sent to radiologist.

The NHS is piloting AI tools in diagnostic imaging, helping radiology teams speed up scan interpretation and reduce backlogs.

Clinical triage tools are being tested to help clinicians prioritise urgent cases more quickly and consistently.

Why it matters for claims

Advances like these shape the evidence we see in future assessments — from better-quality imaging to earlier diagnoses. It means more detailed reports, different types of scans appearing in our files, and evolving expectations around what “best practice” looks like.

AI IN CLAIMS: PAST AND PRESENT APPLICATIONS

Automation in Claims Environments

Thoughtful AI Integration

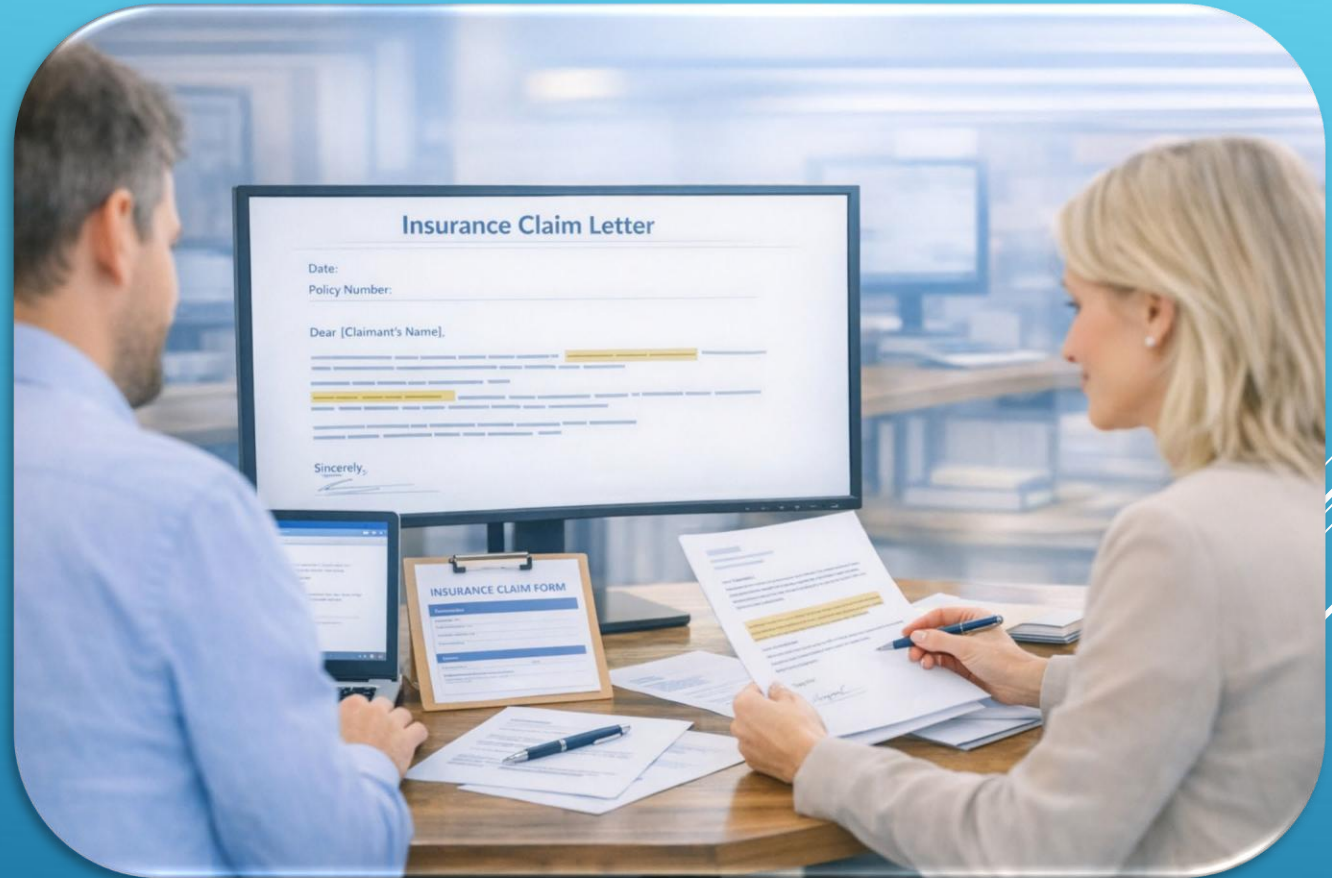
Successful claims automation requires carefully planned AI integration to enhance processes effectively and ethically.

Human Oversight Importance

Maintaining human oversight ensures that automated decisions meet standards and regulatory compliance.

Continuous Performance Monitoring

Ongoing monitoring of AI performance assist's reliability and helps detect and resolve issues promptly.



Introduction of Automation in Underwriting and Claims Processing



Streamlining Routine Tasks

Automation helped simplify repetitive tasks like data entry, increasing operational efficiency and accuracy.

Basic Underwriting Decisions

Automation assisted in making preliminary underwriting decisions faster, reducing manual intervention.

RULES ENGINES

What is it?

A 'rules engine' is a system that executes explicit, human-defined business logic (if/then rules, decision tables, scoring logic) to produce consistent, auditable outcomes.

In insurance, it is primarily used to triage, decide, flag risk, and determine evidence requirements.

They have been used in underwriting applications for 15-20 years (roughly)



Underwriting

Pre-2005 (roughly)

UK protection underwriting relied on:

Paper manuals

Spreadsheet logic

Hard-coded decision trees inside PAS systems

Decisions were rule-based, but:

Logic was embedded in manuals

There was no standalone, configurable URE



Underwriting

Mid-to-late 2000s onwards:

Introduction of tree-based automated decision engines:

Early straight-through processing (STP) for low-risk cases with:

Auto-accept

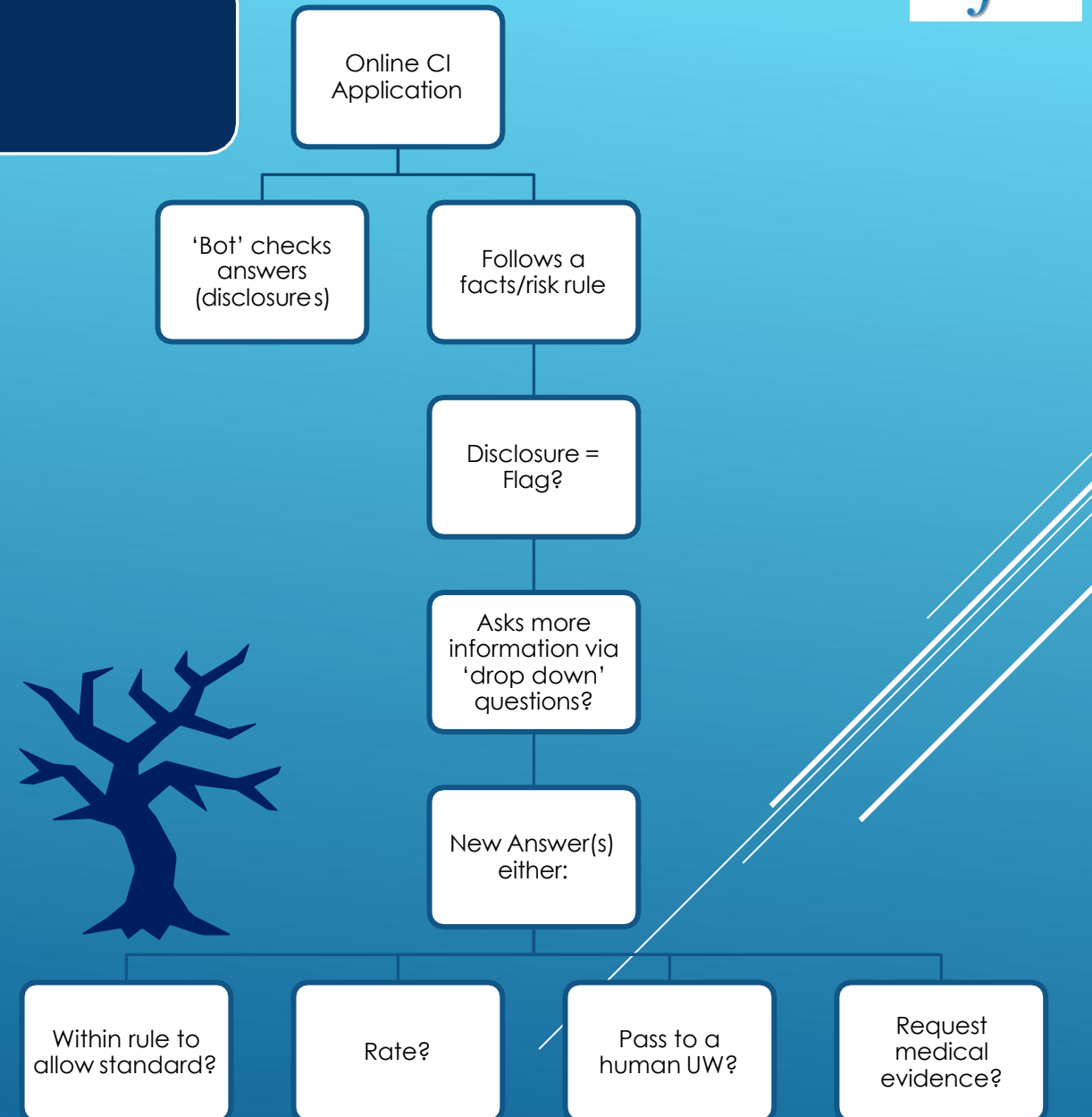
Auto-decline

Refer-to-manual

Rules engines eventually become:

Configurable (not hard-coded)

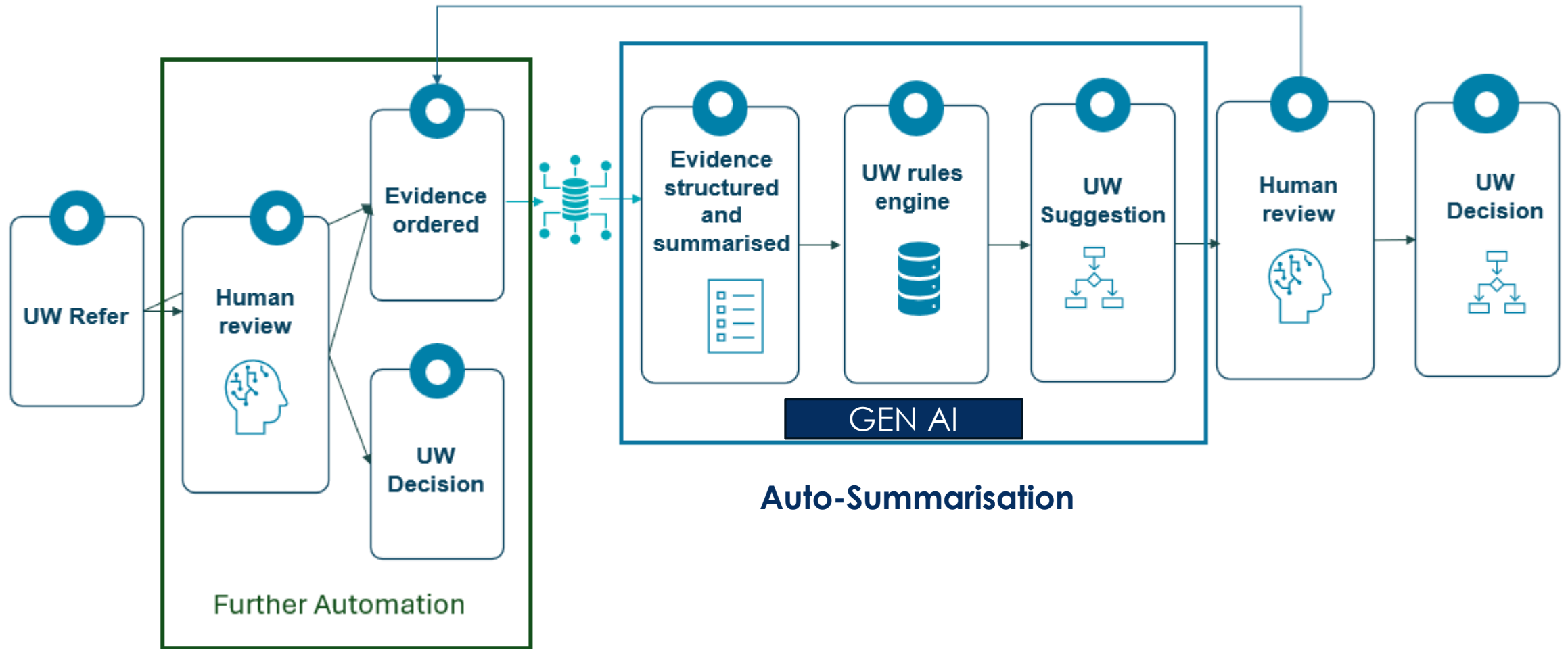
Business-owned rather than IT-owned





AI IN RULES ENGINES

Rules engine evolution: Summarised Evidence Natural Language Generation (NLG)



Claims rules engines



Non-Disclosure and Contestability



Policy Terms and Assessment



Medical Factors



Financial Risks



Return to Work Factors



Occupation Factors

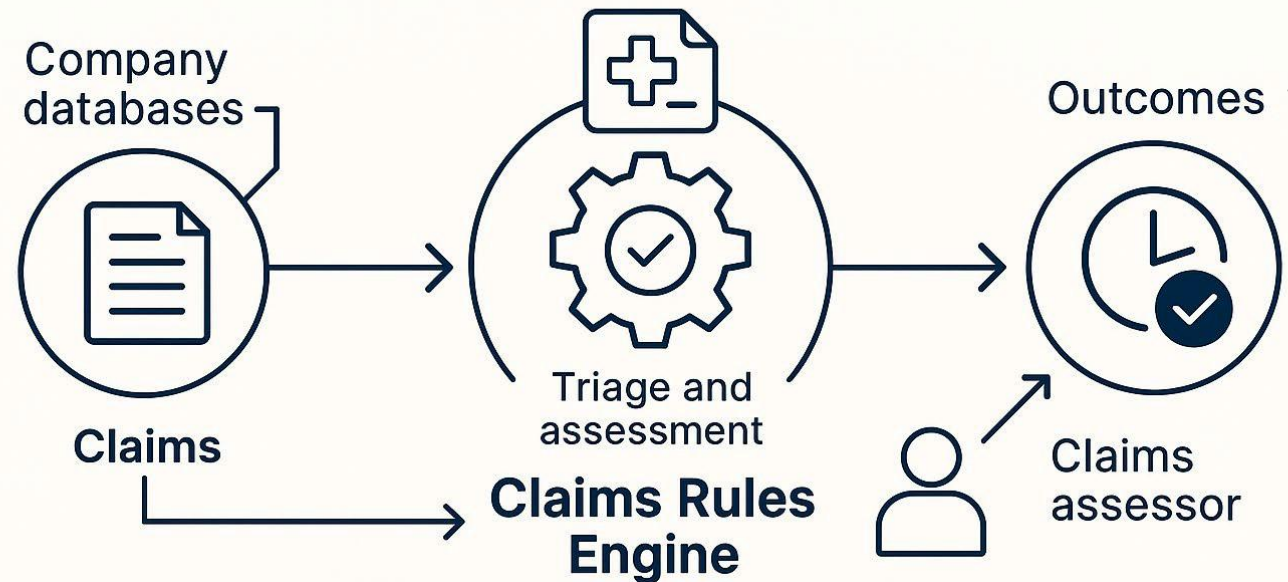


Fraud Mitigation



Jurisdiction and Repatriation

IMPROVED OPERATIONAL EFFICIENCY



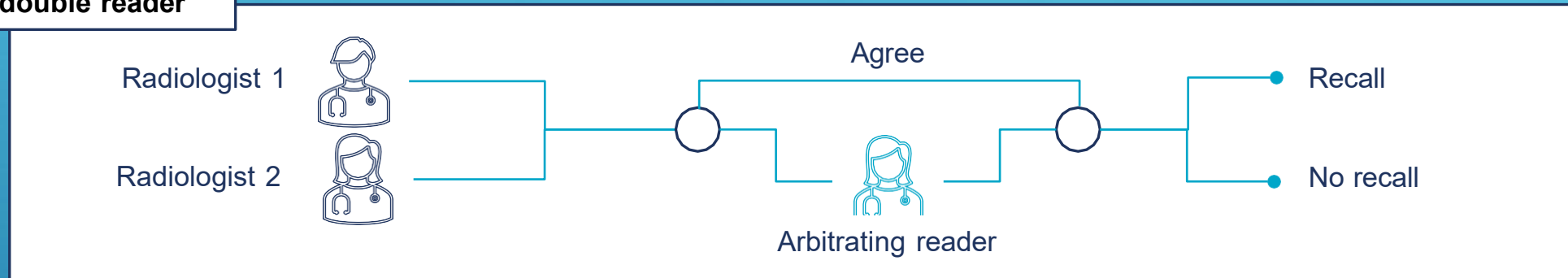
Streamlined workflows, faster processing



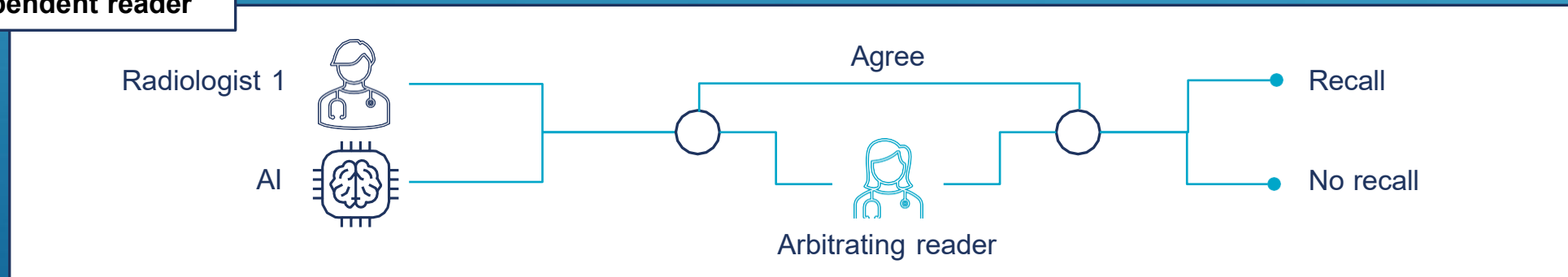
AI IN MEDICAL EVIDENCE

AI IN BREAST SCREENING

Manual double reader



AI independent reader



James Blackwood: Innovate UK research programme based in Scotland, called iCAIRD

BREAST SCREENING

Results from a variety of product vendors whose product is routinely applied to process mammography data



50 %

Reduction in the number of breast radiologists, helping to reduce workforce and recruitment pressures

45 %

Automation of the entire national breast screening process

40 %

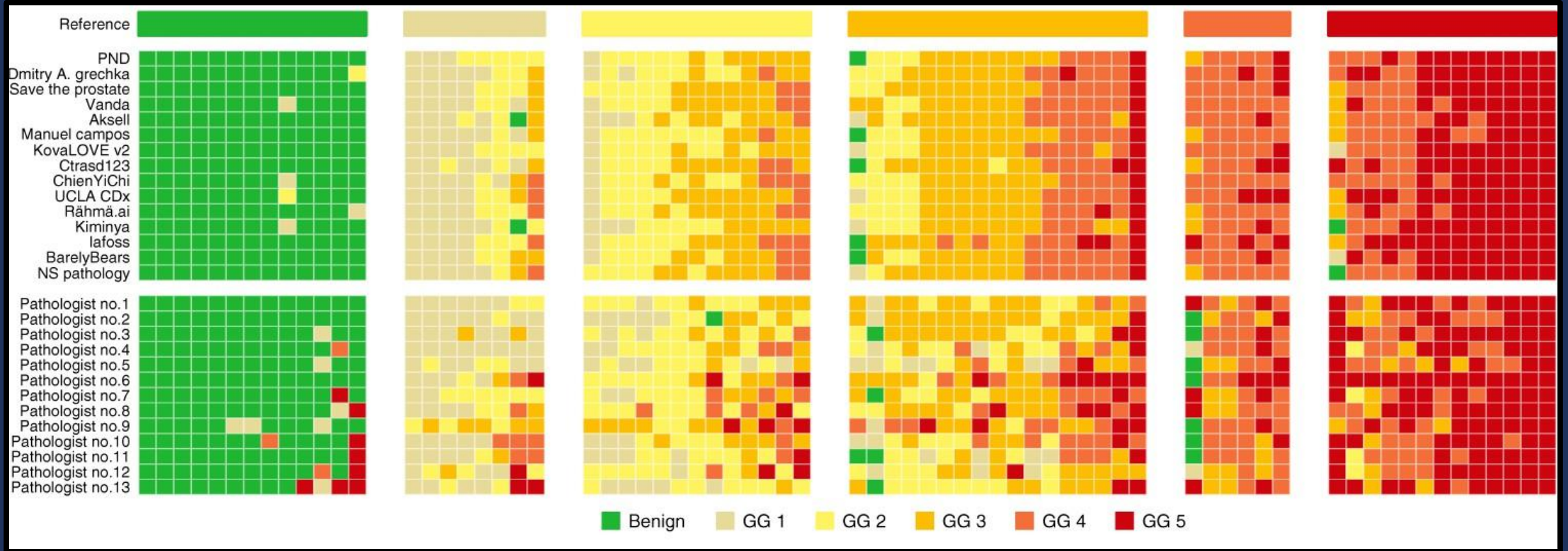
Of breast cancer patients could be diagnosed earlier, some more than 2 years earlier.

12 %

More fatty breast cancers and 5% more dense breast cancers were diagnosed

James Blackwood: Innovate UK research programme based in Scotland, called iCAIRD

ACCURACY: AI vs PATHOLOGIST



Artificial intelligence for diagnosis and Gleason grading of prostate cancer: the PANDA challenge

James Blackwood: Innovate UK research programme based in Scotland, called iCAIRD

A close-up photograph of a person's open palm, facing upwards. A teal-colored awareness ribbon, symbolizing prostate cancer, is draped across the palm and fingers. The background is a solid dark blue.

PROSTATE CANCER

Results from various peer-reviewed journals on the performance of the Paige Prostate product

7.3 %

Increase in the detection of prostate cancer from histopathology slides

65 %

Reduction in patient time-to-diagnosis

70 %

Reduction in false negative diagnoses of prostate cancer

25 %

Improvement in accuracy of Gleason grading

James Blackwood: Innovate UK research programme based in Scotland, called iCAIRD

UTILISING GENERATIVE AI TOOLS IN CLAIMS

What is Generative AI



- ▶ Takes words in and produces new text out (drafts, summaries, explanations).
- ▶ Builds on Narrow AI — but now can generate language instead of following fixed rules.
- ▶ Works through large language models trained on patterns in text.
- ▶ Chat-based tools include ChatGPT, Claude, Gemini, and Microsoft Copilot.
- ▶ Helps with drafting, restructuring, and clarifying — but still needs human sense-checking.

Best Practices: AI as a Supportive Tool, Importance of Sense Checking and Fact Checking



AI as Supportive Tool

AI is there to support claims assessors offering structure, clarity and useful prompts, but it doesn't replace professional judgement or experience.

Sense Checking Importance

Anything generated by AI should still be read with a critical eye. It can be helpful, but assessors need to spot possible errors or inconsistencies. or misses important context.

Fact Checking Necessity

AI can speed up drafting and summarising, but important facts such as medical details, timelines, policy wording, or financial figures should always be verified against the source.

Staying Aware of Limitations

AI works from patterns, not understanding. It may occasionally produce confident but incorrect statements, so it's important to recognise the boundaries of what it can (and can't) do.

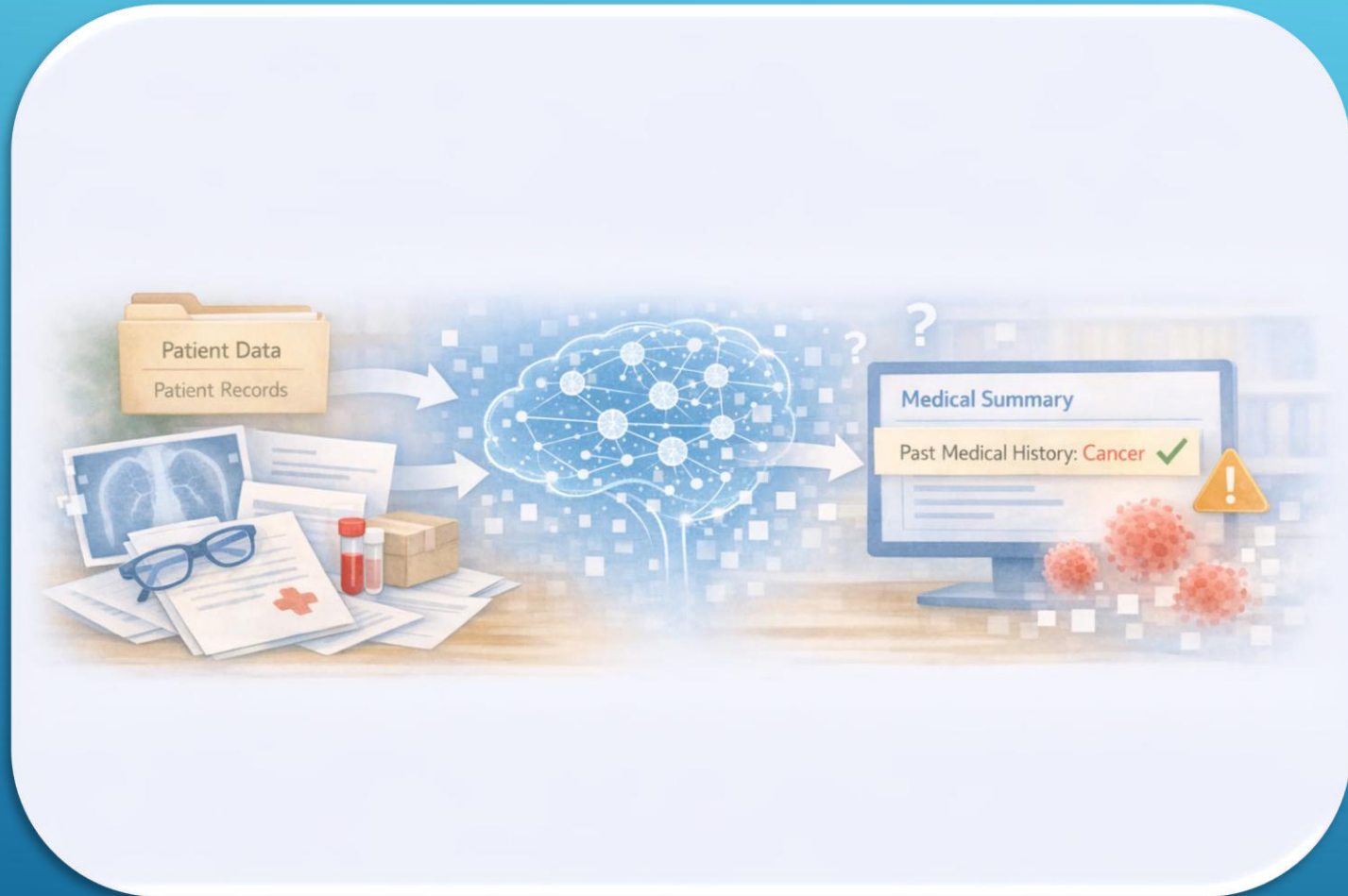
Challenges Brought by AI in Claims: Generated Content and Hallucinated Information

AI Hallucination Risks

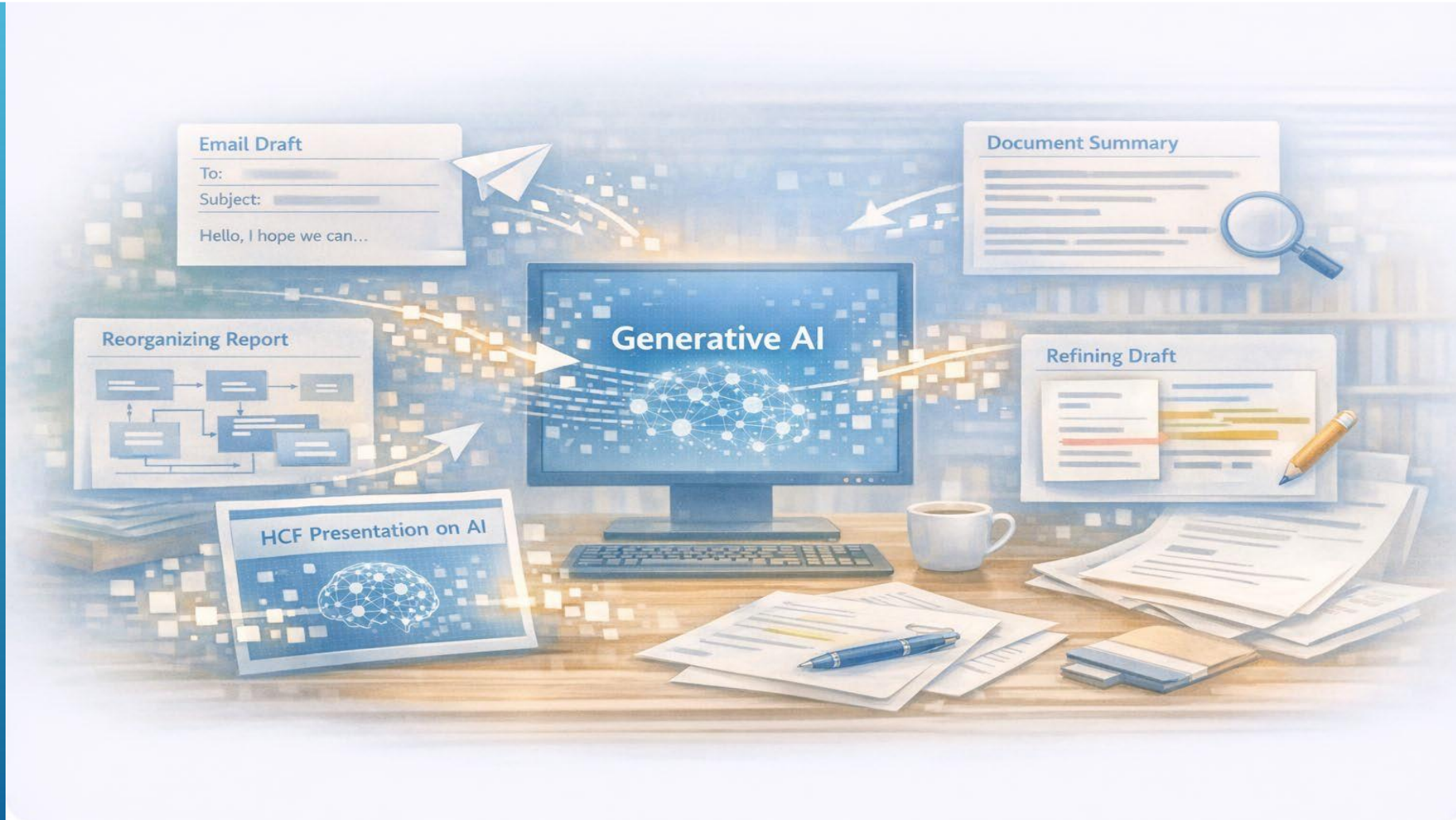
AI-generated content may include inaccurate or fabricated information, leading to potential errors in claims processing.

Impact on Claims Decisions

Hallucinated information can affect the reliability of claim summaries or recommendations, necessitating robust oversight and validation.



How Generative AI Tools Support Claims Work



CHALLENGES IN CLAIMS

Considerations: Data Sensitivity, Regulation, and Access Limitations



Data Sensitivity Importance

Claims data contains sensitive information requiring careful handling to protect individual privacy and confidentiality.

Regulatory Compliance

AI applications must comply with privacy laws and data security standards to avoid legal risks.

Access Controls

Implementing strict access controls ensures only authorised users can access sensitive claims data, maintaining trust.

Appeals



AI is now systemically integrated into society

That means
claimants are
turning to AI for
their appeals

Concern?

No

Increased
consumer
knowledge is
never a bad thing

If our decision was
correct initially, we
should not be
concerned

Does it create
problems?

Yes

increased
appeals causes
operational strain

Appeals – CMO input

No

Increased
consumer
knowledge is
never a bad thing

I have attached a document the claimant sent to the insurer as part of their complaint around negative MRI (it looks AI driven, to me). Does this information look credible?

*“Yes it is.
AI generated it may be, but the
core statements are correct”*

KEY AI CONCEPTS AND TERMS FOR CLAIMS PROFESSIONALS

Natural Language Processing: Understanding Human Language

AI Text Analysis

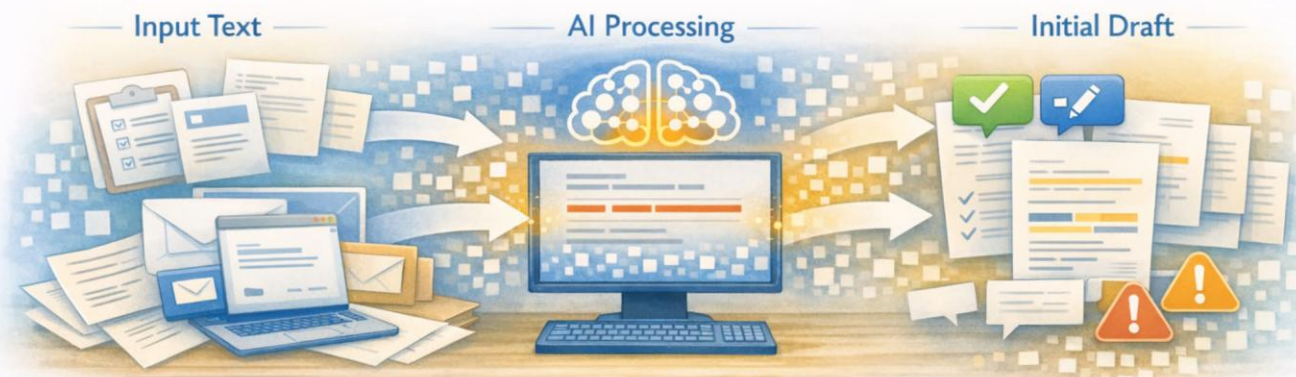
Generative AI uses natural language processing (NLP) to read and interpret complex written information, helping make sense of long claim forms, medical reports, and supporting documents.

Language Generation

NLP also allows AI to produce clear, human-like text — useful when creating first-draft summaries, emails, or explanations that can be refined by the assessor.

Application in Claims and Communication

These capabilities can support claims teams by helping interpret large volumes of documentation and improving the clarity and consistency of written communication.



Large Language Models and Generative AI: How They Work

Training on Large Text Data

Large language models are trained on very large volumes of text so they can recognise patterns, structure, and context in language.

Generating Coherent Content

Once trained, these models can produce coherent, context-aware text — helpful for summaries, explanations, and first-draft writing.

Pattern Recognition, Not Human Understanding

LLMs work by predicting likely words and phrases based on patterns in the data. They support tasks in claims work, but they don't "understand" information in a human sense, so human judgement remains essential.



Agentic AI: Autonomous Agents in Claims and Their Future Potential



What We Mean by Agentic AI

Agentic AI refers to systems that can carry out multi-step tasks independently once set up, following defined goals and rules. These tools can take actions without needing constant prompts, but they still operate within boundaries set by humans.

Potential Use in Claims Workflows

In the future, agentic AI could support more complex claims processes — for example, helping coordinate tasks, gathering information, or progressing routine steps — with the aim of improving efficiency and reducing manual workload.

Why Human Oversight Still Matters

Even with more automation, human oversight remains essential. Claims decisions rely on judgement, context and nuance, so people stay fully in control of outcomes while AI supports the groundwork.

NLP

LLM

Agentic
Sheet + Musicians



CONCLUSION

AI's Transformative Impact

Artificial intelligence improves claims processing through automation and intelligent decision-making.

Understanding AI Concepts

Rules engines will help speed up operational strain

They won't take our job

AI can aid 'clear & warm' communication

Responsible AI Use

AI research is subject to hallucinations – check the source

- Appeals need to be fact checked

- Claims have always made sure adverse decision are regulatory compliant

Personal/Sensitive data usage in AI is company compliant

► Top Tips for Using AI in Claims

- Use AI as a **support tool**, not the decision-maker — it helps you get there faster, but your judgement is still the core.
- **Sense-check** anything it produces. If something feels off, it probably is.
- Keep **sensitive data front of mind** — treat anything you paste in or ask about with the same care as a real case file.
- **Clear prompts = clearer outputs**. A bit of context goes a long way.
- Always **double-check medical facts, dates, policy wording and figures** against the source file before finalising anything.

► Why it matters for claims

- AI makes the admin and first-draft stages lighter, but the final call needs your clinical reasoning, policy knowledge, and experience. These tips keep the efficiency benefits without risking accuracy.

TIPS FOR USING AI IN CLAIMS